



TULSIRAMJI GAIKWAD-PATIL College of Engineering and Technology

Wardha Road, Nagpur - 441108

Accredited with NAAC A+ Grade

Approved by AICTE, New Delhi, Govt. of Maharashtra



(An Autonomous Institute Affiliated to RTM Nagpur University with NAAC A+ Grade)

Department of Information Technology

Session 2025-26(Odd Semester)

VISION	MISSION
To emerge as a learning hub and center of excellence in the domain of Information Technology	[M1] To impart quality technical education through effective teaching learning process. [M2] To provide a platform to address societal issues as well as challenges faced by IT industries. [M3] To foster a culture of research and impart innovative and entrepreneurial skills in the field of IT. [M4] To ensure overall development of students and staff by inculcating knowledge and professional ethics as a part of lifelong learning.

END SEMESTER

Semester: V

Programme: B. Tech Information Technology

Course: Digital Logic & Fundamentals of Microprocessor

Time: 3 Hours

Max Marks: 60 Marks

Instructions to candidates:

1. All questions are compulsory in Group A
2. Solve any two sub-question out of three in Group B
3. Assume suitable data wherever necessary
4. All question carries marks as indicated

BEC33510	Course Outcome
CO1	To classify the working principles of various combinational circuits and their applications in different digital systems.
CO2	To analyze and recognize various sequential components utilized in the design of combinational circuits.
CO3	To develop the skills to design and validate the operation of various combinational and sequential circuits.
CO4	To provide knowledge on the 8085 microprocessor, covering its architecture, functionalities, addressing modes, port types, and practical applications.
CO5	To explore the use of Assembly language programming for the 8085 microprocessor, emphasizing its timing, control features in industrial embedded systems.

Group A				
Que. 1	Answer the following Question	CO	BTL	Marks
a.	Define Boolean algebra.	CO1	2	2
b.	What is encoder?	CO2	2	2
c.	Define race around condition.	CO3	2	2
d.	What is stack in 8085?	CO4	2	2

e.	Define ISR (Interrupt Service Routine).	CO5	3	2
Group B				
Que. 2	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Minimize Boolean function using K-map (5 variables).	CO1	2	5
b.	Explain logic gates with truth tables.	CO1	3	5
c.	Convert SOP to POS form with example.	CO1	2	5
Que 3.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Design a multiplexer using basic gates.	CO2	2	5
b.	Explain arithmetic circuits (adder & subtractor).	CO2	4	5
c.	Design and explain priority encoder.	CO2	3	5
Que 4.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain Master-Slave JK flip-flop.	CO3	2	5
b.	Design synchronous counter.	CO3	3	5
c.	Explain excitation table and its use.	CO3	4	5
Que. 5	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain pin diagram of 8085.	CO4	3	5
b.	Compare different addressing modes with examples.	CO4	3	5
c.	Write assembly program for subtraction.	CO4	4	5
Que. 6	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain interrupt handling process in 8085.	CO5	3	5
b.	Draw and explain timing diagram of opcode fetch cycle.	CO5	4	5
c.	Explain memory interfacing in 8085.	CO5	2	5

Bloom's Taxonomy Level (BTL)- Remember , Understand , Apply, Analyze , Evaluate , Create